E-classroom	**Name:**					
	Gr 10		**Date:**		**Time:**	$2\frac{1}{2}$ **HR**
CAPS Reference	**Function**					
Topic	**Hyperbolic Function**					

1. GOT IT! [15 MIN]

Basic standard:

$$f(x) = \frac{a}{x} + q$$

$f(x)$: – represents the y – value

a: – indicates if the function is positive or negative

x: – indicates the vertical asymptote $x = 0$

q: – indicates the horizontal asymptote $y = q$

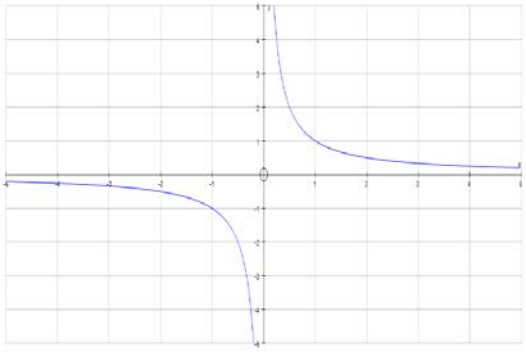
$\therefore q$ –indicates when the function will move up or down

Hyperbola function.

Positive function; $a > 0$

Domain: $x \in (-\infty; \infty)\ and\ x \neq 0$

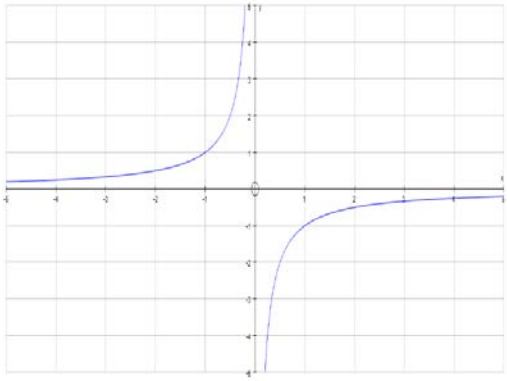
Range: $y \in (-\infty; \infty)\ and\ y \neq q$



Negative function; $a < 0$

Domain: $x \in (-\infty; \infty)\ and\ x \neq 0$

Range: $y \in (-\infty; \infty)\ and\ y \neq q$



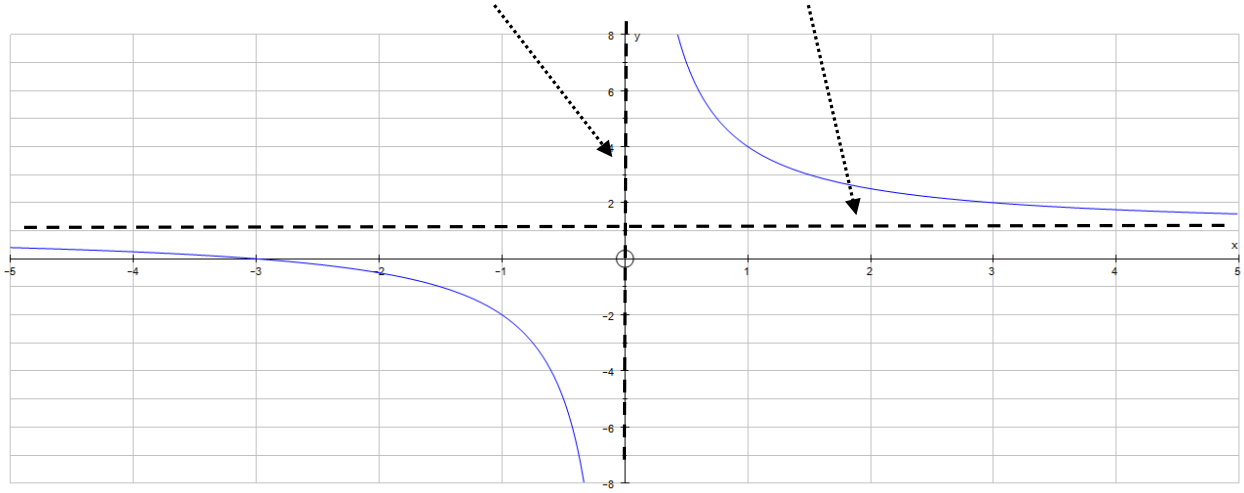
Drawing of a Hyperbolic function.

$$f(x) = \frac{3}{x} + 1$$

Most important!!

Asymptotes

$x = 0$ ***and*** $y = 1$



For the Hyperbola function make use of a table

x	-3	-2	-1	1	2	3
$\frac{3}{x} + 1$	0	$-\frac{1}{2}$	-2	4	$2\frac{1}{2}$	2

Important information about the function.

$a > 0$

This indicates that the function is positive.

Function is drawn in the 1st and 3rd quadrant

$q = 1$

This indicates that the function has moved 1 unit up from the original function $y = \frac{1}{x}$

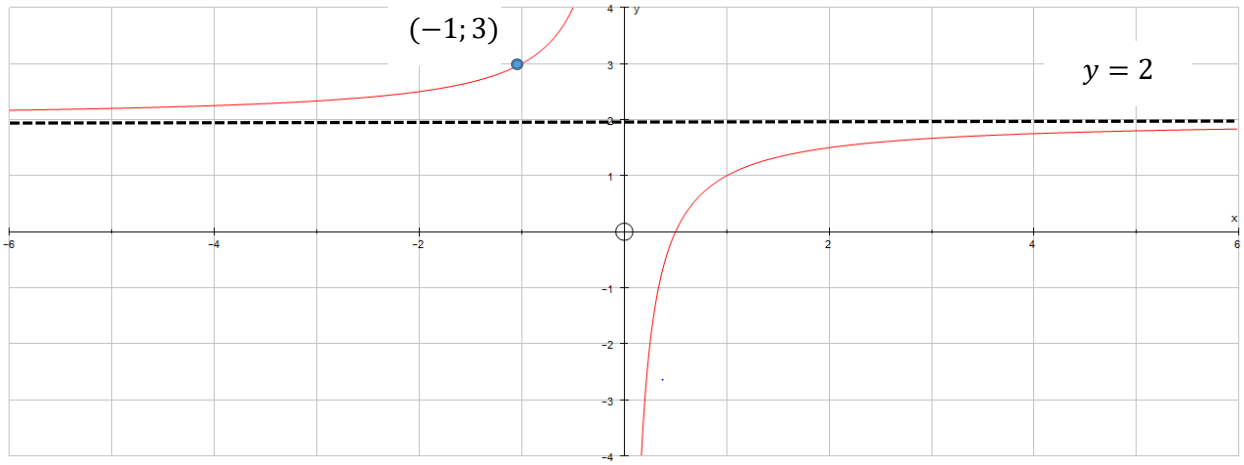
Asymptote $y = 1$. Therefore the function does not touch the line $y = 1$

Domain: $x \in (-\infty; \infty)$ *and* $x \neq 0$

Range: $y \in (-\infty; \infty)$ *and* $y \neq 1$

Determine the equation of a Hyperbolic function

$$\boldsymbol{f(x) = \frac{a}{x} + q}$$



Step 1: Substitute the asymptotes $y = 2$

$$y = \frac{a}{x} + q$$

$$y = \frac{a}{x} + 2$$

Step 2: Substitute the given coordinate $(-1; 3)$

$$y = \frac{a}{x} + 2$$

$$3 = \frac{a}{-1} + 2$$

$$3 - 2 = -1a$$

$$1 = -1a$$

$$-1 = a$$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = \frac{-1}{x} + 2}$$

2. GO AHEAD [60 MIN]

2. Sketch the following on a set of axis and answer the questions that follows.

2.1 $f(x) = \frac{3}{x} - 2$

a) Write down the range of f

b) Write down the horizontal asymptote of f

c) Determine the value of x for which $f(x) < 0$

d) Write down the equation of h, where $h(x) = f(x) + 2$

2.2 $g(x) = -\frac{1}{x} - 1$

a) Write down the range of f

b) Write down the horizontal asymptote of f

c) Determine the value of x for which $g(x) < 0$

d) Write down the equation of h, where $h(x) = -g(x)$

2.3 $f(x) = \frac{4}{x} + 5$

a) Write down the range of f

b) Write down the horizontal asymptote of f

c) Determine the value of x for which $f(x) > 0$

d) Write down the equation of h, where $h(x) = f(x) - 6$

2.4 $g(x) = -\frac{2}{x} + 3$

a) Write down the range of g

b) Write down the horizontal asymptote of g

c) Determine the value of x for which $g(x) > 0$

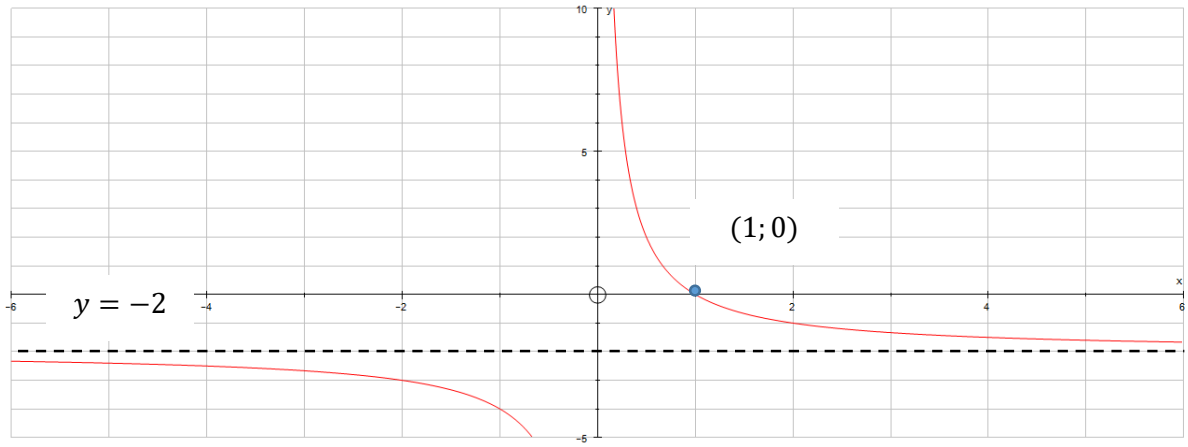
d) Write down the equation of h, where $h(x) = -g(x) + 3$

3. CHECK YOUR WORK![15 MIN]

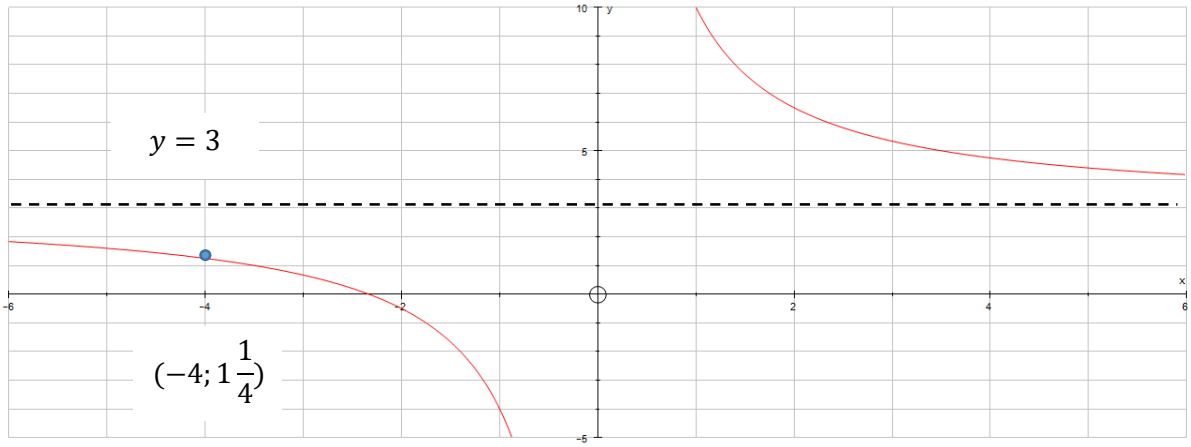
4. GO AHEAD [60 MIN]

4. Determine the equation of the following functions in the format $f(x) = \frac{a}{x} + q$:

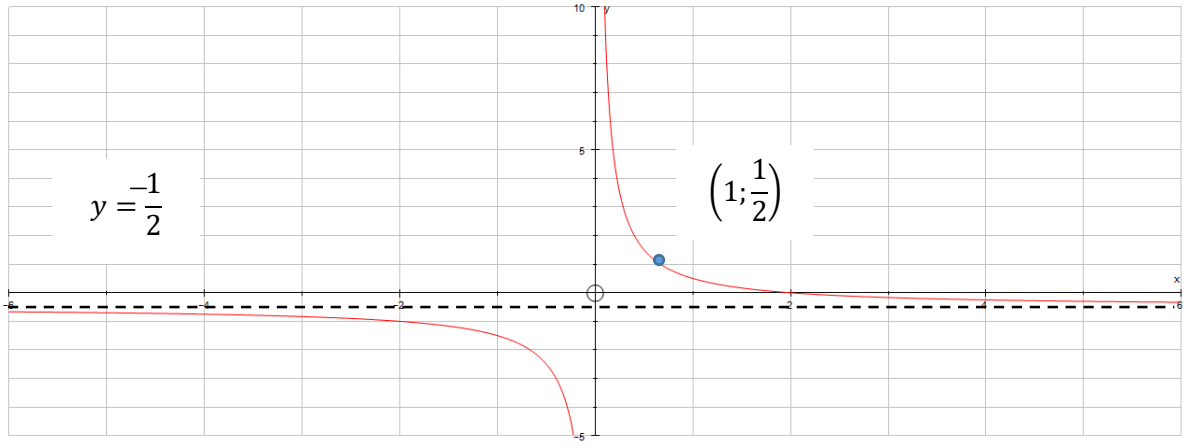
4.1



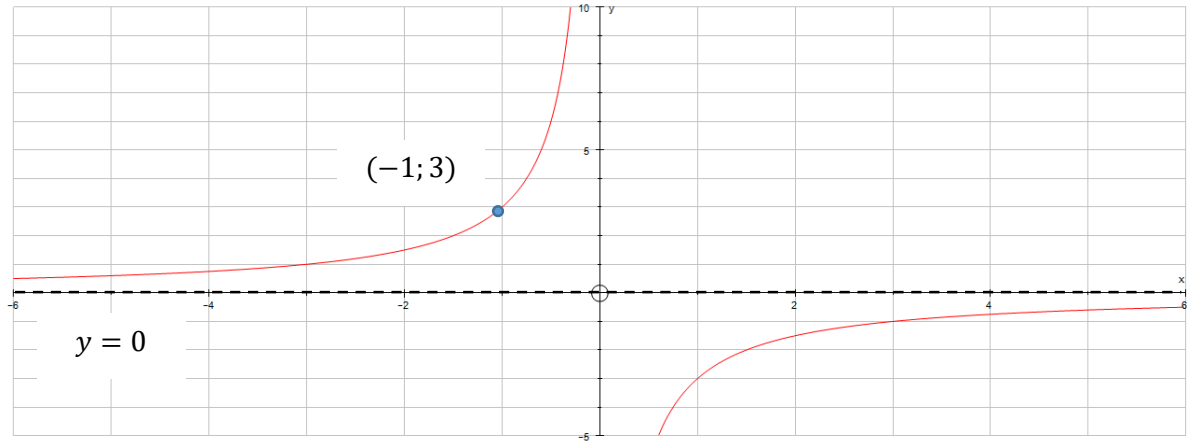
4.2



4.3



4.4



5. CHECK YOUR WORK![15 MIN]